

What is Statistics?

Summarizing Data

Stat 801a

What is statistics?

- Statistics is the science of *learning from data*. It uses the theory of probability to make inferences about populations or processes using data
- For example, we would like to know the mean of a characteristic of a set of elements of a population
- But we can't observe or measure each one because there are too many elements or not all are available.

Examples

- Mean blood pressure of 40 year old pregnant females.
- Mean lifetime of 60-watt bulbs being made at the G.E. plant in Cleveland.
- Mean potency of an antibiotic after storage for 20 months on store shelves.

Statistics is Useful in

- Data collection
 - Sample survey design
 - Experimental design
- Drawing conclusion from data
 - Descriptive methods - summarization and presentation of data
 - Statistical inference - make statements about an entire population based on a sample

Population versus Sample

Since we can't obtain the average exactly, we do the next best thing

- approximate it by finding the average for a *selected subset* of the elements
- we do a statistical study to estimate the population mean
- usually, there are many objectives of a *statistical study*, than just estimating the mean of the population.

Steps in a Statistical Study

- Collecting data - by making observations or taking measurements on some sample or process of interest, possibly as a part of a statistically designed experiment or a survey.
- Summarizing the data - using statistical summary methods (calculating means or constructing a histogram, for example).
- Analyzing the data and making inferences - using models and statistical methods for drawing conclusions about the situation being considered.

Terminology and definitions

- Population - The set of all objects (or measurements) of interest; they are called elements of the population.
- Sample - Any subset of elements from a population.
- A statistical study is made using a sample obtained from a population.
- Sample represents the population.

How to Select a Sample?

Theoretically we can get a random sample of n elements by

- making n draws, one at a time
- on each draw, each remaining element in the population is equally likely to be the one drawn.

Practically we can seldom satisfy the theoretical requirements, so we do the best we can to introduce randomness into our sample selection process.

How to Select a Sample? (continued)

- Can the result of a statistical study of population mean (average) be far away from the true mean? - sure, sometimes. A statistical study provides evidence. It doesn't prove anything
- A random sample selected using the method described earlier ensures that each sample of size n drawn from a population has the same chance of being selected
- Methods that ensure this property are called simple random sampling
- We shall use a method for selecting a simple random sample from a population in a one of our labs using random digit tables

Some Important Definitions

- Data value: measurement
- Observation: Row of data values
- Variable: Set of data values for the same type of measurement
- Data set: The entire collection of data values

Example: Measurements on Mice

Animal id	Cage	Weight	Length
1	1	22.5	5.3
2	1	27.3	4.8
3	1	32.2	4.5
4	2	23.7	4.9
5	2	30.9	5.7
6	2	22.6	5.2

Data Description

- Two broad applications of Statistics
 - descriptive statistics
 - inferential statistics

- When measurements from an entire population is available to us data description will be a major objective.
- In this case, methods for organizing, summarizing, and describing data are needed.
- Good descriptive statistics enable us to make sense of the data by reducing large amounts of data to a few summary measures and graphical displays.

Data Description - cont.

- When only a random sample is available from a large population (or a process), inferential statistics becomes the major focus.
- However, even in this case, descriptive statistics of the sample data is still important as they can be used to draw conclusions about the population from which the sample was taken.
- Data descriptions are of two types:
 - graphical methods
 - numerical summaries

Graphical Methods

- Bar Chart
- Histogram
- Stem-and-Leaf plot
- Time Series plot
- Boxplot
- Normal Probability Plot

Numerical Summaries

- Measures of Center or Location
 - Mode
 - Median
 - Mean
- Measures of Variability or Spread
 - Range
 - Percentile (Quantile, Quartile)
 - Interquartile Range
 - Variance
 - Standard Deviation
 - Coefficient of Variation

Histograms

- To plot a histogram, a *frequency table* must first be constructed
- The range of the data is divided into *class intervals* or *bins*
- The number of such intervals is determined by the end-points of the intervals *number of observations* in the data set
- The larger the number of observations the more the number of class intervals we may use
- Divide the range of data values by the number of class intervals to determine an approximate *class width*

Weight gains of chicks fed an antibiotic - Data

An animal scientist is carrying out an experiment to investigate whether adding an antibiotic to the diet of chicks will promote growth over the standard diet without the antibiotic. The scientist determines that 100 chicks will provide sufficient information to validate the results of the experiment. From previous research studies, the average weight gain of chicks fed the standard diet over an 8-week period is 3.9 grams. The scientist wants to compare the weight gain of the chicks in the study to the standard value of 3.9 grams. The weight gains for the 100 chicks are recorded in the table shown on the next slide.

Example: Weight gains of chicks fed an antibiotic

3.7	4.2	4.4	4.4	4.3	4.2	4.4	4.8	4.9	4.4
4.2	3.8	4.2	4.4	4.6	3.9	4.1	4.5	4.8	3.9
4.7	4.2	4.2	4.8	4.5	3.6	4.1	4.3	3.9	4.2
4.0	4.2	4.0	4.5	4.4	4.1	4.0	4.0	3.8	4.6
4.9	3.8	4.3	4.3	3.9	3.8	4.7	3.9	4.0	4.2
4.3	4.7	4.1	4.0	4.6	4.4	4.6	4.4	4.9	4.4
4.0	3.9	4.5	4.3	3.8	4.1	4.3	4.2	4.5	4.4
4.2	4.7	3.8	4.5	4.0	4.2	4.1	4.0	4.7	4.1
4.7	4.1	4.8	4.1	4.3	4.7	4.2	4.1	4.4	4.8
4.1	4.9	4.3	4.4	4.4	4.3	4.6	4.5	4.6	4.0

Creating a Histogram

- It was decided to have 10 class intervals
- Since the range is $4.9 - 3.6 = 1.3$, a class interval width of $1.3/10 \approx .1$ was used
- The end-points of the intervals are incremented by one-half the unit of measurement (i.e., .05 gram, here)
- This is so that no observation falls on any end-point
- Since the smallest observation is 3.6 we start with the interval $3.55 - 3.65$
- Once the class intervals have been determined observations in the data set falling into each of these classes are tallied

Reading in the chick data

```
chick.tbl <- read.table(text = "
      3.7  4.2  4.4  4.4  4.3  4.2  4.4  4.8  4.9  4.4
4.2  3.8  4.2  4.4  4.6  3.9  4.1  4.5  4.8  3.9
4.7  4.2  4.2  4.8  4.5  3.6  4.1  4.3  3.9  4.2
4.0  4.2  4.0  4.5  4.4  4.1  4.0  4.0  3.8  4.6
4.9  3.8  4.3  4.3  3.9  3.8  4.7  3.9  4.0  4.2
4.3  4.7  4.1  4.0  4.6  4.4  4.6  4.4  4.9  4.4
4.0  3.9  4.5  4.3  3.8  4.1  4.3  4.2  4.5  4.4
4.2  4.7  3.8  4.5  4.0  4.2  4.1  4.0  4.7  4.1
4.7  4.1  4.8  4.1  4.3  4.7  4.2  4.1  4.4  4.8
4.1  4.9  4.3  4.4  4.4  4.3  4.6  4.5  4.6  4.0")
```

```
chick <- unlist(chick.tbl, use.names = F)
head(chick, n = 10)
```

```
[1] 3.7 4.2 4.7 4.0 4.9 4.3 4.0 4.2 4.7 4.1
```

Frequency table for the chick data

Frequency Table

```
cuts <- cut(chick, breaks = seq(3.55, 4.95, by = 0.1))
freq <- table(cuts)
freq
```

```
cuts
(3.55,3.65] (3.65,3.75] (3.75,3.85] (3.85,3.95] (3.95,4.05] (4.05,4.15]
      1           1           6           6          10          11
(4.15,4.25] (4.25,4.35] (4.35,4.45] (4.45,4.55] (4.55,4.65] (4.65,4.75]
     13          10          13           7           6           7
(4.75,4.85] (4.85,4.95]
      5           4
```

Relative Frequency Table

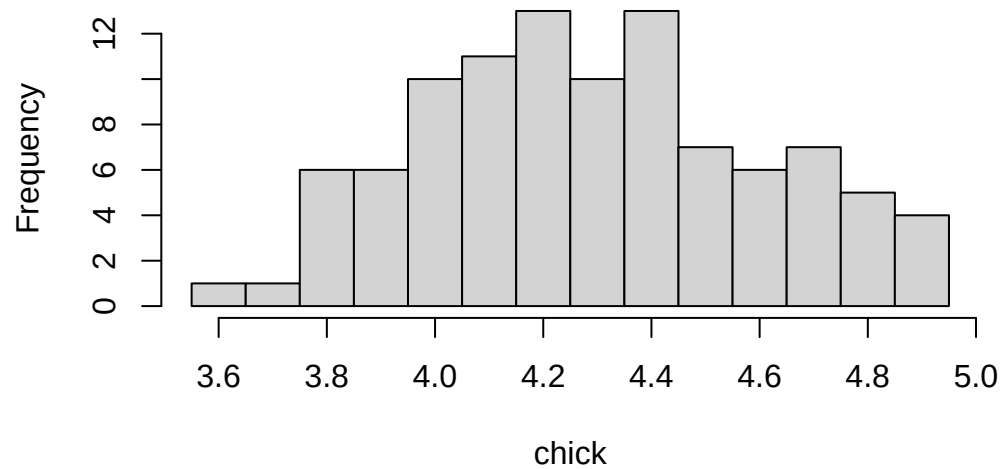
```
rel_freq <- table(cuts)/length(chick)
rel_freq
```

```
cuts
(3.55,3.65] (3.65,3.75] (3.75,3.85] (3.85,3.95] (3.95,4.05] (4.05,4.15]
      0.01       0.01       0.06       0.06       0.10       0.11
(4.15,4.25] (4.25,4.35] (4.35,4.45] (4.45,4.55] (4.55,4.65] (4.65,4.75]
     0.13      0.10      0.13      0.07      0.06      0.07
(4.75,4.85] (4.85,4.95]
     0.05      0.04
```

Creating a Histogram (base R)

```
hist(chick, breaks = seq(3.55, 4.95, by = 0.1))
```

Histogram of chick



Creating a Histogram (ggplot2)

```
# install.packages(tidyverse)
library(ggplot2)
ggplot(mapping = aes(x = chick)) +
  geom_histogram(binwidth = 0.1, color = 'black', fill = '#1a80bb')
```